

ABOUT YOU KAY AI FASHION ASSISTANT

Technical Architecture

TECHNICAL ARCHITECTURE — INTRODUCTORY NOTE

This document presents a comprehensive three-phase technical architecture for KAY AI Fashion Assistant, designed to evolve About You's chat support into a conversational fashion discovery platform.

The architecture is built on a foundation of proven, industry-standard technologies selected for their production maturity, scalability, and ecosystem support. However, the specific technology stack presented here (cloud infrastructure, database, backend framework, ML infrastructure, and LLM provider) represents our recommended approach and is subject to alignment with About You's existing technical infrastructure, business agreements, and organizational technology preferences.

Core Principles:

- Phase progression is independent of specific tool choices
- API-first integration with About You's existing services
- Minimal re-architecture of About You's core systems
- Clear separation of concerns (KAY logic vs. commerce platform)

The technology recommendations serve three purposes:

1. Demonstrate technical feasibility of the proposed product roadmap
2. Provide a baseline for conversations with About You's technical teams
3. Enable rapid prototyping during Phase 1 MVP development

All final technology decisions will be made collaboratively with About You's engineering and infrastructure teams to ensure optimal integration, cost efficiency, and alignment with organizational standards.

ABOUT YOU KAY AI FASHION ASSISTANT

Technical Architecture

Phase 1 — MVP: Conversational Discovery

Timeline: 0–6+ weeks

Scope: E1–E7

What we build

A core conversational shopping experience where users describe what they need naturally, KAY understands the request, retrieves relevant products, explains why they match, and supports the path to Add to Bag and purchase through ABOUT YOU's existing commerce flow.

Technology stack

- **Frontend:** React + TypeScript
Used for the KAY chat interface, product cards, preferences, and mobile interaction.
- **Backend:** Node.js + TypeScript
Handles APIs, routing, session/context management, and integration with ABOUT YOU services.
- **AI / LLM:** GPT-class or Claude-class LLM
Used for intent understanding, extracting preferences and constraints, clarification questions, fashion guidance, and recommendation explanations.
- **Search:** PostgreSQL full-text/product search for the initial MVP implementation
- **Ranking:** Rule-based ranking using signals such as relevance, product quality, availability, and popularity
- **Database:** PostgreSQL
Stores product metadata, conversation context, and session data.
- **Integration:** ABOUT YOU APIs
Product catalogue, pricing, availability, product details, Add to Bag, and existing checkout.
- **Cloud:** Google Cloud Platform
Example services: Cloud Run and Cloud SQL.
- **Analytics:** BigQuery
Tracks recommendation interactions, Add-to-Bag events, purchase attribution, CSAT, and funnel metrics.

Key capability

User describes need → KAY understands intent/context → Search products → Rank relevant items → Explain recommendations → View Product → Add to Bag → Existing checkout → Buy

ABOUT YOU KAY AI FASHION ASSISTANT

Technical Architecture

The MVP is mainly **LLM + search + rules**, rather than a complex trained recommendation model.

Phase 2 — Enhanced Recommendations + Image Discovery

Timeline: Months 2–4

Scope: E8–E10

What we add

Phase 2 improves recommendation quality and introduces multimodal discovery and deeper shopping integrations.

New technologies

- **Machine Learning Ranking:** XGBoost or LightGBM
Learns from behaviour such as product clicks, Add-to-Bag events, purchases, skips, and recommendation feedback.
- **Computer Vision:** CNN / vision models such as ResNet
Used to analyse uploaded fashion images.
- **Multimodal Embeddings:** CLIP or similar embedding model
Converts product images and uploaded inspiration images into comparable vectors.
- **Vector Database:** Pinecone, Weaviate, or equivalent
Enables similarity search across fashion-product embeddings.
- **Image Similarity Search:**
Finds products visually similar to an uploaded photo.
- **Body Scan integration:** Existing ABOUT YOU fit/body capabilities accessed via API
- **Virtual Try-On integration:** Existing VTO capability connected into the KAY journey
- **Account integration:** KAY can use permitted ABOUT YOU account context where relevant.

Technologies retained from Phase 1

React · Node.js · LLM · PostgreSQL · ABOUT YOU APIs · GCP · BigQuery

Key capability

**Phase 1 + Upload image → Analyse visual features → Find similar products
→ ML improves ranking using behaviour → Offer Body Scan/VTO where relevant**

ABOUT YOU KAY AI FASHION ASSISTANT

Technical Architecture

At this stage KAY starts moving from **rules-based recommendations toward learned ranking**.

Phase 3 — Persistent Personalization + Voice

Timeline: Months 4–8+

Scope: E11–E13

What we add

Phase 3 turns KAY into a more persistent, personalized assistant that can learn from the shopper over time.

New technologies

- **Personalization / Collaborative Filtering:**
Learns preferences from interactions, clicks, purchases, favourites, recommendation feedback, and similar-user behaviour.
- **User Style Profile:**
Stores consented preferences such as size, colours, styles, brands, budget and other long-term signals.
- **Cross-session personalization:**
KAY can use previous interactions instead of treating every conversation as completely new.
- **Voice:** Whisper or another speech-to-text model
Converts spoken shopping requests into text.
- **Multimodal interaction:**
Combines text, images and potentially voice within the same session.
- **Advanced personalized ranking:** Neural ranking/recommendation models
Ranks products specifically for the individual shopper rather than only by general relevance.
- **Forecasting / proactive recommendations:** Time-series or predictive ML where justified
Can identify patterns and potentially surface relevant recommendations proactively.
- **Notifications:** Push-notification infrastructure for consented proactive experiences.

Technologies retained

React · Node.js · LLM · PostgreSQL · Vector Database · ABOUT YOU APIs · GCP · BigQuery · ML ranking infrastructure

Key capability

ABOUT YOU KAY AI FASHION ASSISTANT

Technical Architecture

Phase 1 + Phase 2 + persistent user preferences → personalized ranking → multimodal/voice interaction → increasingly tailored fashion assistance

The main shift in Phase 3 is from:

“What does this user want right now?”

to:

“What does this user usually prefer, and how should that context improve this shopping session?”

Technology evolution in one line

Phase 1:

LLM + Search + Rules → prove conversational discovery

Phase 2:

+ Machine Learning + Computer Vision + Vector Search → improve relevance and enable image discovery

Phase 3:

+ Persistent Personalization + Advanced Ranking + Voice → evolve into a more personalized AI fashion assistant

****KEY QUESTIONS TO ASK ABOUT YOU****

Infrastructure:

1. "What's your primary cloud provider—GCP, AWS, or Azure?"
2. "What's your standard database? PostgreSQL?"
3. "Do you have a preferred backend language or framework?"
4. "What analytics platform do you use?"

Integration :

5. "Are the SCAYLE product APIs the right integration point for KAY?"
6. "How would KAY authenticate and integrate with your backend?"
7. "Should KAY have isolated data storage or shared database?"

ML Infrastructure :

ABOUT YOU KAY AI FASHION ASSISTANT

Technical Architecture

8. "Do you have ML training and serving infrastructure we can use?"
9. "Any preferences on LLM provider—OpenAI or Claude?"